FALL 2024

CSE-4355 Electromechanical Systems and Sensors

THE UNIVERSITY OF TEXAS AT ARLINGTON



LAB 2 REPORT

Submitted toward the partial completion of the requirements for CSE 4355

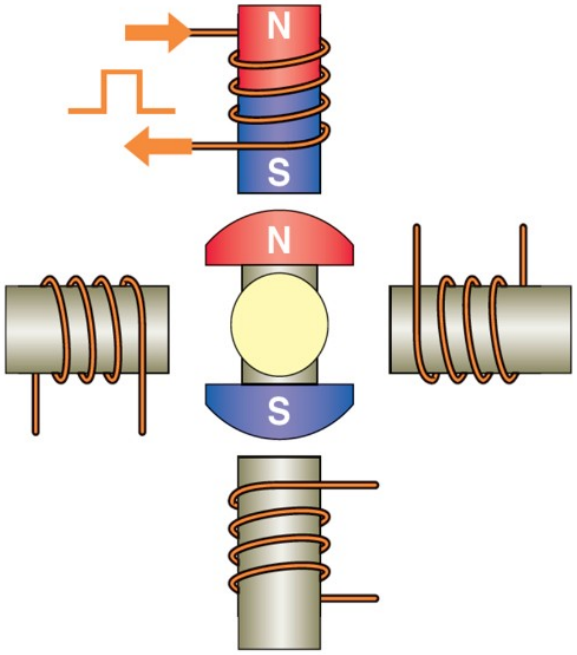
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Introduction:



A stepper motor is a type of motor that moves in discrete steps or increments, unlike traditional motors that spin continuously. Stepper motors move by powering coils in a sequence, creating a magnetic field that attracts a motor, in our case a permanent magnet. By controlling which coils are energized and when, the motor can be made to step forward or backward in precise increments. In this lab, we will be familiarizing ourselves with this motor and attempt to micro step between these discrete steps to have the motor move more smoothly. This will be done by changing the magnetic field strength by changing PWM with the cosine function.

Findings:

After levelling out the balance bar and connect coil A to a 10.4V power supply. We noticed that the balance beam is now hard to turn. Using the hanging weight, we slid the weight on the beam until the motor cannot create enough torque to keep the beam from moving. We measured the mass of the weight as 225 grams and the distance to be 12 centimetres. Lowering the voltage to 9.4 made the distance lower to about 11 centimetres. Lastly, lowering the voltage to 8.4V and repeating the previous steps we noticed that lowering the voltage lowers the current, current and torque are directly related. Meaning, as the current was lowered, there was less torque from the motor. This made it so that there would be less force needed to move the motor. The following graph shows our findings at the different voltages: 10.4 being the top, 9.4 middle, and 8.4 bottom.

Weight of object(g)	distance from center(m)	Gravity constant	Torque = F * D
225	0.12	9.81	264.87
225	0.11	9.81	242.8
225	0.105	9.81	231.76
F = weight * gravity			

Powering the motors in a pattern of : (+ -)(* *), (* *)(+ -), (- +) (**), (**) (- +) we saw that the motor turned clockwise. Reversing this pattern allowed us to move in the opposite direction or counterclockwise. However, if we wanted to incorporate micro-stepping, we need there to be in-between states of where both magnets are powered at the same time but by varying strengths. This was done by approximating the unit circle in the magnetic field in and passing that ratio between sine and cosine to power coils A and B. This allowed us to have the motor be more precise. The pattern would then have been changed to:

(+ -)(* *), (+ -)(+-) ,(* *)(+ -), (+ -)(-+),

(- +) (**),(- +)(- +), (**) (- +), (+ -)(-+)

Results:

Each full rotation of a stepper motor is divided into a series of steps, allowing for precise control of movement, which makes them ideal for applications where accuracy and control are essential. Micro stepping allows us access to the in-between steps of the motor. In our case, we have a motor that steps ins 1.8 degrees increments but with micro stepping we can step in lower increments, such as 0.9. Micro-stepping significantly reduces the vibration that can occur with standard full-step operation and provides smoother movement and increased range of the motor. This is done by driving the motor in smaller increments, the torque variations associated with full stepping are minimized, leading to more consistent performance. Overall, micro stepping enhances the performance and applications of stepper motors, making them suitable for precision applications where smooth movement and accuracy are crucial.

Issues:

Some issues that came up were often intermediate rounding done by the program of which was unintended and sometimes the program skipping a single step when switching directions. The control algorithms and driving circuitry was much more complex than the simpler single step driving and caused many troubleshooting issues. We also had many issues with things outside the scope of the lab such as passing the information through UART.

Circuit Built:

